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SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I A Althaf (192521257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Experiments

QUESTIONS:4 **Total Marks:40**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies 2	Good understanding with proper integration of most concepts 2	Basic understanding; limited or partial integration 2	Poor understanding; unable to integrate concepts 1
Experimental Design	30	Well-planned, systematic implementation with optimal solution 2	Correct implementation with minor issues 2	Basic implementation with errors or inefficiencies 2	Incorrect or incomplete implementation 1
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration 2	Appropriate use of tools with correct execution 2	Limited or inconsistent use of tools 2	Incorrect or minimal use of tools 2
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results 1	Good analysis with reasonable interpretation 2	Basic analysis with limited insights 2	Poor or incorrect analysis of results 2
Documentation	10	Clear, well-structured documentation with diagrams, code, and results 1	Organized documentation with necessary details 1	Basic documentation with missing details 2	Poor or incomplete documentation 2

8 9 10 8

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding, some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

...re packets during a TCP connection setup

i) Aim:

To capture and analyze the TCP three-way handshake using wireshark and understand how the SYN, SYN-ACK, and ACK packets establish a reliable TCP connection

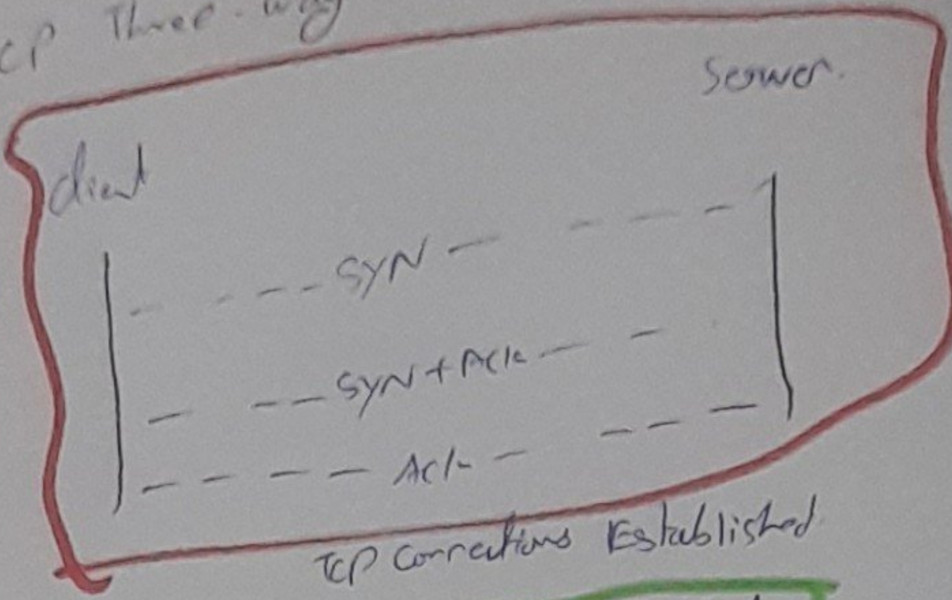
ii) Requirement

- Computer/laptop
- Internet connection
- wirshark
- web browser
- TCP protocol

iii) Procedure

- open wirshark.
- select the active network interface such as wifi
- click start capturing method.
- open a web browser and access website.
- stop the capture after the connection is established
- Apply the following diplay filtered.
- Identify the TCP packets involved in the connection established.
- select the packets and examine the TCP flags. Sequence number and Acknowlegment number field.

iv) TCP Three-way Handshake



v) How Reliability is Established:

TCP uses sequence number and acknowledgement to provide reliable communication. During the handshake, both side synchronize their sequence number and confirm that communication is possible. The final Ack confirms both devices are ready to exchange data.

vi) Result

The TCP three way handshake was successfully identified using wireshark. The SYN-Ack, syn and ACK packets demonstrated how TCP established a connection and synchronize both communication devices.

... a DNS query using UDP in wireshark

i) AIM:-

To capture a DNS query using UDP in wireshark compare the UDP header size with TCP header, losing on header size, sequence number, acknowledgement, speed and reliability.

ii) Requirement:-

- Computer/Laptop
- Internet connection.
- wireshark
- DNS service
- UDP protocol.

iii) Procedure:-

i) open wireshark.

ii) select the active network interface such as wifi or Ethernet.

iii) start packet capture

iv) open a web browser.

v) Enter a domain such as google.com

vi) stop the program.

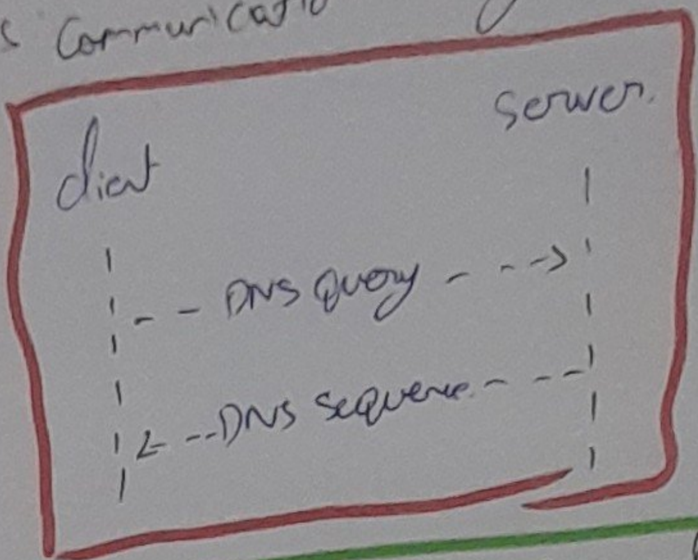
vii) Apply the following wireshark filter: DNS

viii) select a DNS query packet.

xi) Expand the UDP datagram. protocol (UDP)

x) observe the source port, Destination port and checksum.

iv) DNS Communication using UDP:



v) Absence of sequence number and acknowledge.

Unlike TCP, UDP does not contain sequence numbers or acknowledgement number field in its header therefor, UDP does not itself confirm whether a packet has reached the destination.

vi) Result:

The DNS query was successfully captured using wirshark. The UDP was observed to be 8 bytes compared with minimum 20-byte. The header. UDP does not contain sequence number or acknowledgement number and therefore has lower alead.

...mulate packet loss and observe TCP

Retransmission vs UDP.

i) AIM

To simulate packet loss and analyses the behaviour of TCP and UDP using wireshark, especially how TCP retransmits lost packets while UDP does not provide transmission.

ii) Requirements

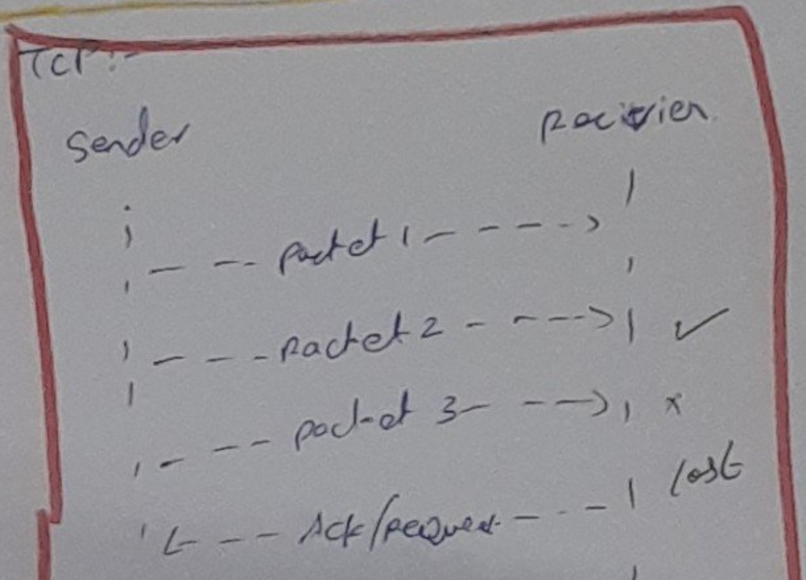
- computer/laptop.
- wireshark
- Internet/network
- TCP
- UDP

iii) Procedure

- open wireshark.
- select the active network interface such as wifi
- start capturing packet.
- start a network activity that generates TCP
- Reconnect the network.
- Stop the wireshark capture.

- Apply the following filter to observe TCP retransmission: tcp.analysis.retransmission
- Examine the TCP packets before and after the interruption
- Capture the UDP traffic separately and observe whether the same TCP-style transmission reaction occurs.

iv) Packet Loss Scenario:



v) Result

The packet-loss experiment demonstrates that TCP retransmits lost packets, whereas UDP does not provide TCP-style transmission. TCP is therefore more reliable but has greater overhead.

DNS Traffic capture and Resolution Time anal.

i) AIM :-

To capture DNS traffic using wireshark, identify the DNS query type and response measures the DNS resolution time and analyse why DNS commonly used UDP

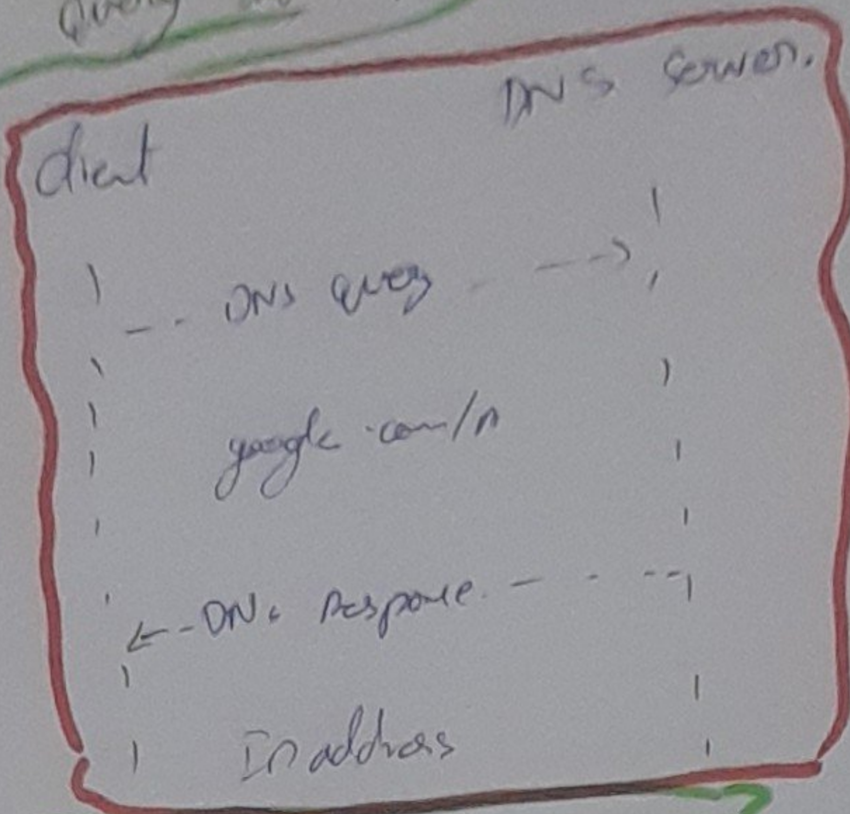
ii) Requirement:

- computer /laptop.
- Internet.
- wireshark
- webbrowser
- DNS service.

iii) Procedure

- open wireshark.
- select the active network interface such as wifi
- select Packet capture .) open a web browser.
- Enter a domain name file for. eg. google.com.
- stop the capture after the domain is required.
- Apply the wireshark filter : DNS.
- Identify the DNS query packet
- Record the query type, requested domain, response

iv) DNS Query and Response:



v) Why DNS Typically uses UDP:-

DNS commonly uses UDP because DNS queries and response are usually small & do not require a TCP connection to be established first.

vi) Result:

The DNS traffic was succesfully captured using wireshark. The DNS query type such as AAA, AAAA or MX and its corresponding response were identified. The DNS resolution time was calculated from one to another.